



Thoughts on population dynamics

An internal discussion paper for Wye Salmon Association (WSA)

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Introduction

Atlantic salmon numbers have been falling in the Wye for several decades, broadly as they have for the whole of England and Wales and, indeed, for the wider North Atlantic population. This note attempts to cut through to some fundamental mathematical truths around population dynamics and hopefully dispel some myths which may distract from the actions most likely to have a useful impact.

I'm not claiming any expertise in this area, I'm not a biologist, so I may get some of the detail wrong (most of what follows I've gleaned from Wikipedia). But, as a physicist, I was taught the utility of simple models in trying to understand system behaviour. The life cycle of the salmon spans both a highly complex river ecosystem and a highly complex marine ecosystem. Consequently, it's very easy to get lost in the detail and miss the basics. And possibly ending up twiddling the wrong knobs.

Mathematics of growth

Exponential

Unconstrained growth is exponential. The rate of change of population is proportional to the population (P) multiplied by a growth factor (r). It's that simple.

$$\frac{dP}{dt} = r \cdot P$$

Equation 1: differential form of the exponential equation.

This is the equation describing the initial growth of everything from bacteria to elephants (and people). For the record, the resulting differential equation is:

$$P(t) = P_0 \cdot e^{rt}$$

*Equation 2: first order differential equation;
P₀ being the initial population.*

But this form is not often required in the modern world of Excel; equation 1 is all you need. We can rearrange equation 1 to calculate discrete, incremental population cycles, as per equation 3.

$$P_{n+1} = (1 + r) \cdot P_n$$

*Equation 3: incremental form as coded in Excel;
n is the reproductive cycle*

This is easily set up in Excel and copied down a few lines to generate the graphs below.

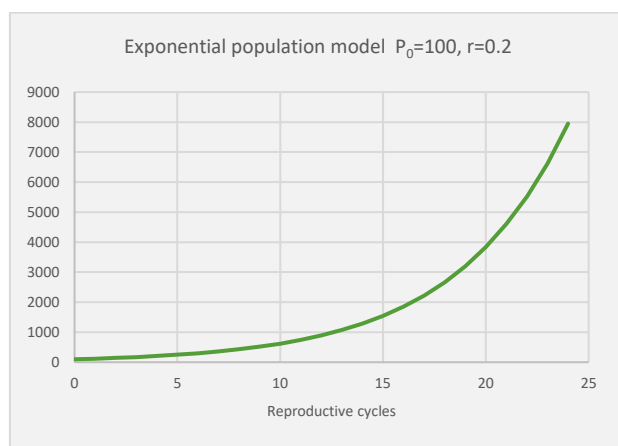


Figure 1: rising population with $r > 0$

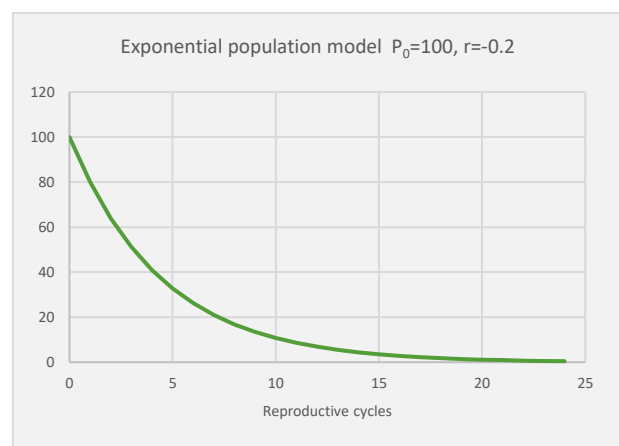


Figure 2: falling population with $r < 0$

Note that if $r > 0$, we get exponential growth and if $r < 0$, we get exponential decay. You may recall the discussion of r values during the covid pandemic (the bit Boris apparently couldn't get his head around!). If k was exactly equal to 0, we would get a stable situation. But no biological systems are ever that precise.

Logistic

In the real world, exponential growth is limited by finite resources such as food and habitat. A common model used by ecologists which incorporates this idea is known as the logistic model. In this, a “carrying capacity” term K is added, as shown in equation 4.

$$\frac{dP}{dt} = r \cdot P \left(1 - \frac{P}{K} \right)$$

Equation 4: differential form of the logistic equation;
 K is the carrying capacity of the available habitat.

Now, the exact form of the term in brackets appears somewhat arbitrary, but this simple form apparently provides a useful approximation for many situations. For completeness, the corresponding differential equation is:

$$P(t) = \frac{K}{1 + A \cdot e^{-rt}}$$

Equation 5: differential form of the logistic equation;
the constant A is derived from initial condition by

$$A = \frac{K}{N_0} - 1$$

And the incremental form is:

$$P_{n+1} = P_n \left[1 + r \left(1 - \frac{P_n}{K} \right) \right]$$

Equation 6: incremental form as coded in Excel;
 n is the reproductive cycle

Running this in Excel with different starting conditions gives us the chart shown in figure 3.

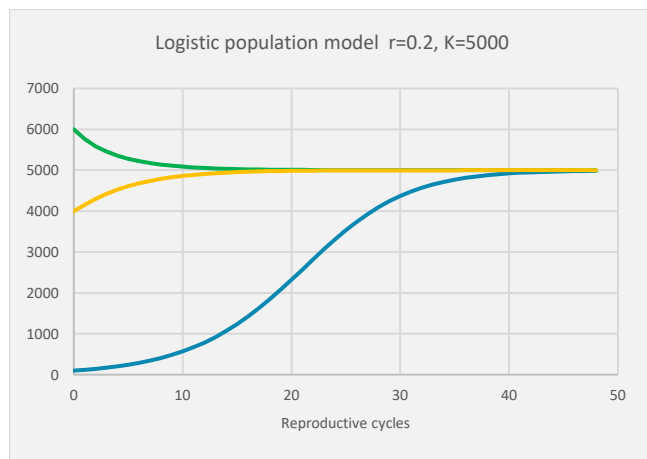


Figure 3: logistic population development from different starting conditions. $K=5000$ and $r=0.2$ throughout. The blue curve shows growth starting at 100; up to around cycle 15 it is essentially identical to the exponential model in figure 1 but diverges beyond this, due to the carrying capacity term. The orange and green curves start from a population of 4000 and 6000 respectively.

Comparing the blue curve to figure 1, we can see that below about 30% of carrying capacity, the logistic growth curve closely matches the exponential growth curve. Above this point it progressively flattens out as it approached the carrying capacity. The other two curves essentially show us what happens when carrying capacity changes. Imaging K changed from 6000 down to 5000, then the green curve would be the response. Likewise, the yellow curve could be caused by K changing from 4000 to 5000. As long as r is positive, population tracks the carrying capacity.

To emphasise this point, because it is important, figure 4 shows what happens when we take a stable population and alter K . Figure 4 commences with the population in equilibrium at a carrying capacity of 5000. From cycle 5 onwards, K is reduced to 4000 and, after a few cycles, the population establishes a new equilibrium. After cycle 30, K is set back to 5000, and the population progressively increases accordingly. At no point does the population go into terminal decline. As long as r is positive, changes in carrying capacity result in a new equilibrium population.

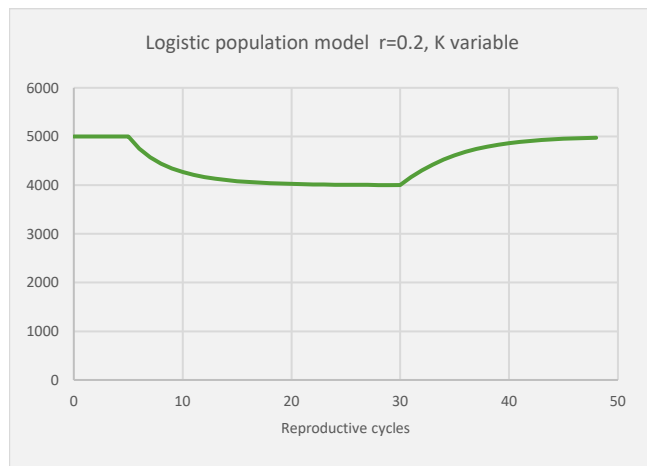


Figure 4: logistic population development starting with $K = 5000$; at cycle 5, we reduce K to 4000 and, at cycle 30, set it back to 5000.

But what happens when r goes negative? Figure 5 shows us. With an intrinsic r value of 0.2, the population sits at the carrying capacity. When, at cycle 10, we impose a reduction in adult survival rate from 1.0 to 0.8, the population goes into exponential decay. In this scenario, we have a healthy carrying capacity, but the population falls because each successive generation is smaller than the last. The dashed blue curve in figure 5 shows the response if we double the carrying capacity after cycle 20. Essentially, it makes almost no difference to the outcome. When a population has already fallen significantly due to negative r , increasing carrying capacity will not change the outcome.

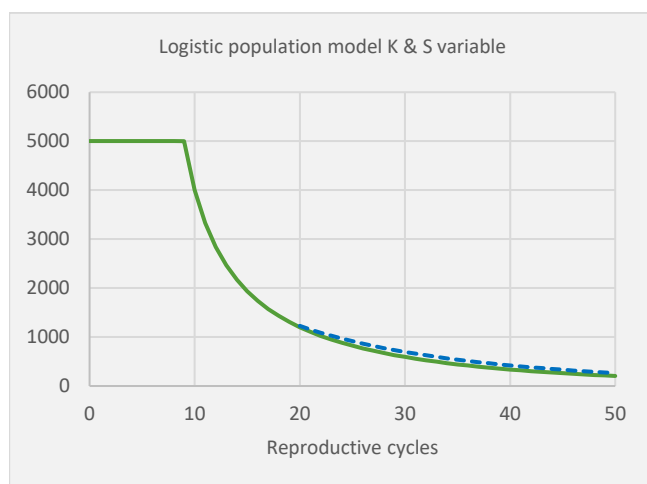


Figure 5: logistic population development following a sudden crash in s from 1.0 to 0.8 from cycle 10. The result is an exponential decay.

The dashed blue curve shows what happens if K is increased from 5,000 to 10,000 after cycle 20.

Variability

The above is a very idealised treatment, producing smooth population transitions. But obviously the real world is not like that. It is common for fish population to have large swings in population due to a variety of factors. K may be modulated by availability of food species, weather conditions, etc. Also, r may be modulated by predator numbers (both in the freshwater and marine phases) and so forth. Indeed, it is common for predator-prey relationships to oscillate wildly in what is termed a dynamic equilibrium. Thus, many species are found to “bounce around” beneath the carrying capacity to some degree.

Let's next try this out by adding a random variation in survival rate. If we take equation 6 and multiply the right-hand side by a random number varying between 0.7 and 1.0, we simulate this random variability. The result is shown in figure 6. The population gets knocked back by low survival years but, crucially, it recovers strongly in the better years. It never quite gets up to the carrying capacity but bounces around beneath it. Despite some alarming dips, this is a stable long-term situation.

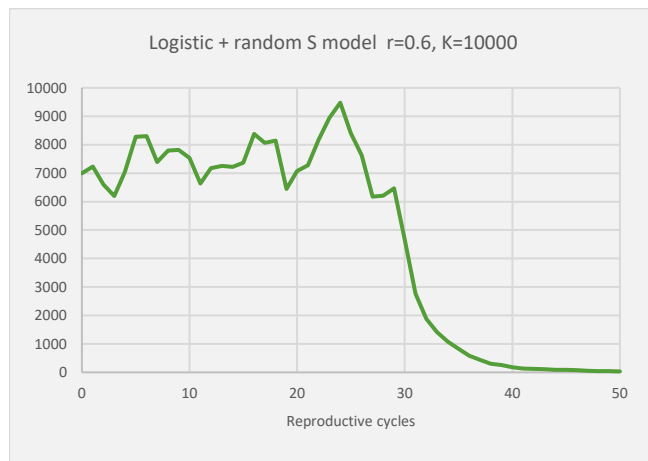


Figure 6: logistic population development with added random survival rate S where $0.7 < S \leq 1.0$. In this model, the population never reaches the carrying capacity but oscillates randomly beneath it. Between cycles 29 & 30, we reduce S ($\times 0.6$), causing the exponential decline.

The above analysis is entirely generic, taking no account of the specifics of the salmon lifecycle. However, there are already some lessons to draw from this simple model.

Relevance to Wye salmon

The AST population modeller

The [Atlantic Salmon Trust](#) website is explicit that, for UK rivers, the carrying capacity applies primarily to parr. In their [salmon population modeller](#), the relationship between eggs and parr is highly non-linear, peaking out at the carrying capacity. This is shown in figure 7.

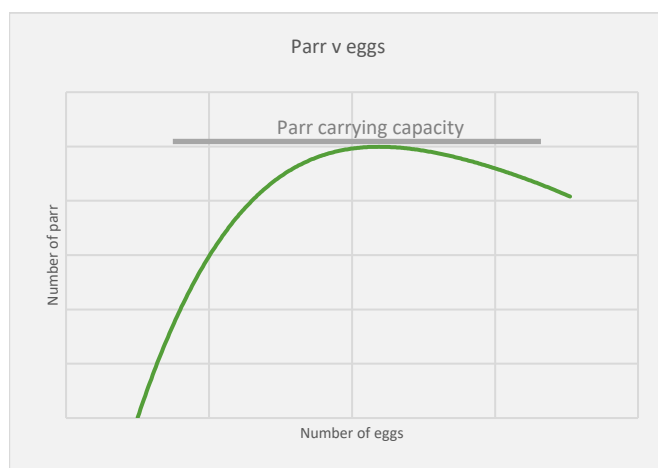


Figure 7: non-linear relationship between number of eggs and number of parr.

For low populations (i.e., low egg numbers), the gradient is 13.4%. Meaning that 13.4% of eggs produce a successful parr. As the population grows, egg number increase, taking us up the curve, which progressively reduces the recruitment factor; as the population reach the carrying capacity, this has reduced to 2.2%. Should egg numbers increase further still, the system responds by further reducing egg to parr survival. Under no circumstances can parr numbers exceed the carrying capacity of the available habitat. In practice the parr capacity is set by their aggressive territoriality.

There is a further non-linear stage between parr and smolts, as shown in figure 8. However, this stage never hits a carrying capacity, since this was already established at parr stage. Nevertheless, the recruitment factor falls off significantly as numbers increase. For low populations, the recruitment factor is 50%. At the parr carrying capacity, this has fallen to 25%.

The nett effect of the “density dependent mortality” in these two life stages serves to stabilise the adult population in the face of year-to-year variation in egg numbers. But note that the maximum density is not a universal constant; rather, it varies by location, driven by local food abundance, availability of cover, etc.

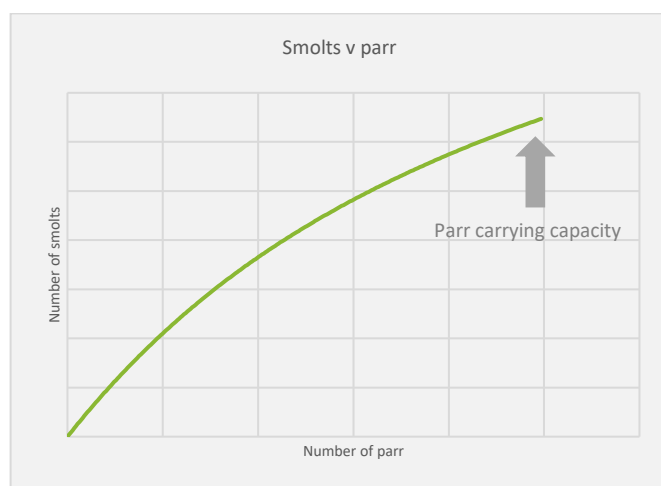


Figure 8: non-linear relationship between number of parr and number of smolts.

Other survival factors in the AST model are linear. A hen deposits a fixed number of eggs and a fixed fraction of smolts return as adult salmon, etc. Of note is that marine survival, which historically was circa 25%, is said now to be circa 5%, with 7% appearing to be the default value in their modeller.

Applying some basic numbers

Much is made of the salmon's resilience which results from its "massive" overproduction of eggs. However, this safety margin turns out to be decidedly finite. Since the Wye is not an index river, as a starting point, I'm going to use some population statistics from the Tamar, which is. Recent data, based on fish count data plus genetic testing, has been reported in King et al., 2023 and is summarised in table 1 below.

	Fraction of population	Of which, females	Average eggs per female
1SW fish	68%	46%	3700
2SW fish	32%	80%	7670

Table 1: basic population statistics from the Tamar.

I shall make use of John's input that, at best, 30% of returning females make a viable redd, plus the above recruitment/survival numbers from the AST modeller. We begin with 100 spawning salmon and calculate each life stage population. Using the numbers in table 1, within this population we have 31 1SW hens and 26 2SW hens which together carry 312,088 eggs.

Scenario 1: pristine river, well below carrying capacity, with historic marine survival rate

Carried eggs		312,088
Eggs deposited in viable redds	30%	93,626
Parr	13%	12,546
Smolt	50%	6,273
Returning adults	25%	1,568
Returning adults which spawn	80%	1,255
r value		11.5

Table 2: calculation for the case of a small, developing population in a pristine river, with good marine survival.

The historic situation did indeed have a good margin. 100 first generation salmon produced over 1200 successful progeny: $r = 11.5$. But this growth doesn't go on for ever. After a few generations, the population hits the carrying capacity. The next table shows this calculation.

Scenario 2: pristine river, at carrying capacity, with historic marine survival rate

Carried eggs		312,088
Eggs deposited in viable redds	30%	93,626
Parr	2.2%	2,060
Smolt	25%	515
Returning adults	25%	129
Returning adults which spawn	80%	103
r value		0.0

Table 3: calculation for the case of a small, stable population in a pristine river, with good marine survival.

In this scenario, the population has stabilised at the carrying capacity by winding back parr and smolt recruitment. Note the substantial reduction in parr and smolt numbers. Our 100 first generation salmon produced approximately 100 successful progenies: $r = 0$. But the original r value of 11.5 is still “in reserve” should any future stressors emerge.

Scenario 3: pristine river, well below carrying capacity, with current marine survival rate

Carried eggs		312,088
Eggs deposited in viable redds	30%	93,626
Parr	13%	12,546
Smolt	50%	6,273
Returning adults	7%	439
Returning adults which spawn	80%	351
r value		2.5

Table 4: calculation for the case of a small, developing population in a pristine river, with current marine survival.

This is similar to scenario 1, but with the current marine survival according to AST. The r value is greatly reduced but still positive. The population would again grow (more slowly) to the carrying capacity and remain stable. But the original safety margin is now greatly reduced by the large drop in marine survival. Any substantive, additional stressor could easily tip the r value into negative territory. Enter the goosander.

Scenario 4: well below carrying capacity, with parr predation & current marine survival rate

Carried eggs		312,088
Eggs deposited in viable redds	30%	93,626
Parr	13%	12,546
Smolt	10%	1,255
Returning adults	7%	88
Returning adults which spawn	80%	70
r value		-0.3

Table 5: calculation for the case of a small, developing population in a pristine river, with current marine survival and, additionally, significant parr predation.

In this final scenario, we retain the current 7% marine survival rate and add in a factor of 5 reduction in parr to smolt recruitment – the result of parr predation. This is not fanciful. In our analysis of e-fishing data (Green, 2026), we found a factor of 10 reduction in parr survival on the Chwerfri, for example. The combined effect is to take r negative, meaning exponential decay and inevitable extinction. The original pristine situation in scenario 1 could withstand the fall in marine survival, or the increase in parr predation in isolation, but not both together.

Wye rod-catch returns

The Wye, not being an Index River, lacks any concrete data on population. This being the case, we fall back on rod-catch data as illustrated in figure 9.

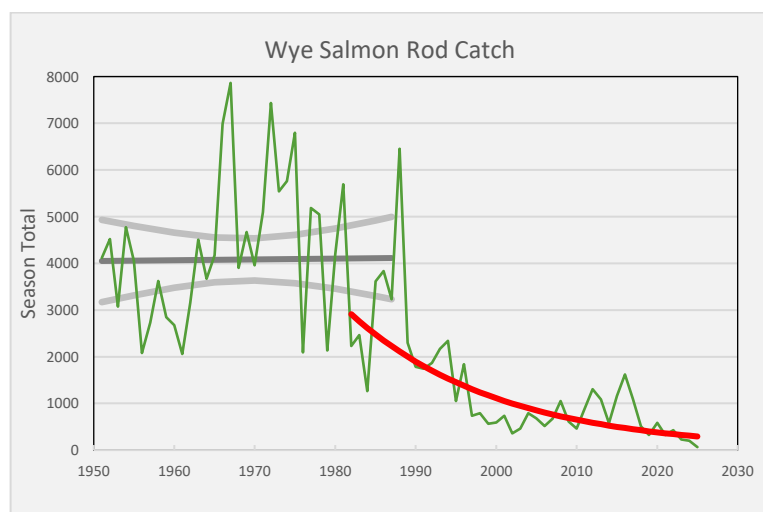


Figure 9: Wye salmon rod-catch. The regression line (dark grey) suggests a stable underlying trend from 1950-88, despite considerable variation caused by year-to-year changes in overall survival. The light grey curves represent the 90% confidence range on the regression.

The red curve is a regression on the period 1982-2024, using an exponential function.

Looking at the catches from 1950 to 1988, annual returns vary between 2000 and 8000, with a mean value of 4000. There are undoubtedly many factors at play driving this variability but, nevertheless, the underlying long-term trend was steady during this period. The linear regression to this period (in grey) supports this interpretation. An immediate conclusion is that the freshwater carrying capacity during this period was most probably circa 10,000, stated as rod-catch. Assuming a 14% exploitation rate, the actual carrying capacity would be about 71,000 returning adults over that period.

Beyond the mid to late 80s, the population went into rapid decline. There is still a proportional amount of year-to-year variability in the chart, but the underlying trajectory is clear. A crucial point is that there is no suggestion of a new, lower equilibrium, as illustrated in figure 4. Thus, it is highly implausible that this decline has been driven by a fall in habitat capacity. In fact, there is a strong qualitative similarity to figures 5 & 6, where the decline is driven by a negative r value.

Wye r value

Developing this idea further, the red curve in figure 9 is an exponential regression applied to the period 1982-2024. The equation of this best-fit exponential is:

$$P = A.e^{-0.053.t}$$

Equation 6: an exponential regression to the rod catch data from 1982 to 2024. P is population (i.e. rod-catch), A is a constant and t is time in years.

Comparing to equation 2, we see that -0.053 is the r value for the population, per year. Over a 4-year reproductive cycle (for simplicity, ignoring multi-SW fish), the r value is -0.21; in other words, an inter-generational survival rate of 79%. This is not theoretical modelling; the number falls directly out of the rod-catch data.

Habitat

The assertion here that the fall in Wye salmon numbers is not driven by habitat loss is supported by our analysis of NRW e-fishing data (Green, 2026) which showed that salmon are still spawning in most of the usual places in the headwater streams, just in lower numbers. In some cases, such as the Llynfi tributaries, “lower” meaning zero. There is no evidence in the e-fishing data to support a thesis of significant habitat loss since the 1980s. Rather, the evidence shows a massive collapse in parr survival throughout large areas of the upper Wye catchment commencing during the 1980s and remaining substantially constant since then.

Moreover, today’s rod-catch numbers of circa 200 represent 2% of our estimate of the 1950-88 rod-catch carrying capacity (10,000). This implies that the salmon population is in the true exponential region, well away from carrying capacity influence. Figure 5 showed very clearly what happens when we increase carrying capacity in such circumstances: essentially nothing. If r is negative, no amount of tinkering with extra habitat will turn the situation around.

By this token, barrier removal – say the fish pass on the Monnow weir at Osbaston – laudable as it is, cannot possibly help the overall Wye salmon population. Accepting that a few fish have indeed been seen in the upper reaches, two comments come to mind. Firstly, the outlook for their progeny is no different to anywhere else in the catchment. If r remains negative, they will go extinct. Secondly, the fish which did go up the Monnow would otherwise have spawned in their natal stream elsewhere in the catchment. In the absence of positive r , this is just robbing Peter to pay Paul. As worthwhile as barrier removal is in a general sense (salmon are not the only species which would like to move up and down rivers), if projects like this are justified primarily on the basis of helping the salmon, this is surely a poor use of scarce funds.

Conclusion

- We have examined the basic behaviour of the logistic population model. Two key parameters emerge from the mathematics.
 - Carrying capacity, K , determines the maximum stable population.
 - Growth factor, r , determines whether the population is growing or declining. If r is positive, the population grows, ultimately limiting at the carrying capacity.
- Achieving or maintaining a population at its carrying capacity requires positive r .
- K will only affect population development for populations which are at or close to the carrying capacity. If the population is well below this level, the value of K has no impact.
- A clear corollary of this is that if r is negative, expanding habitat area (thereby increasing K) will not drive an increase in the population. Expanding a population into new habitat absolutely requires positive r .
- The AST salmon population modeller and its supporting material reinforces the above analysis.
 - For salmon, the carrying capacity is primarily set by density dependent mortality at the parr stage. Parr to smolt recruitment also has a density dependence but is not the primary limit.
 - Thus, “it is not possible to increase the number of parr in a river above its carrying capacity”.
 - Interstage recruitment numbers taken from the AST modeller show that under pristine conditions, with historic marine survival rates, salmon lay circa 12x more eggs than are required to maintain the population (r value 11.5).
 - As the population reaches the carrying capacity, density dependent mortality dramatically reduces egg to parr survival to bring the nett r value to zero. The parr/smolt stages effectively become a “flow regulator” on the system.
 - AST suggest that marine survival has fallen to around a fifth of what it used to be. A factor of 5 takes a huge chunk out of the initial 12x egg surplus, leaving a much smaller buffer against further stressors.
- The e-fishing data shows a very significant, sudden drop in parr survival across much of the upper Wye catchment, commencing in the 1980s. Combined with the fall in marine survival, this would be expected to take r negative.
- The Wye rod catch data is a good qualitative match for such a collapse in r , commencing in the 1980s. It is not a good match for falling K , and nor does the e-fishing data support such a hypothesis. An exponential regression to the rod-catch data from 1982 to 2024 indicates that the intergenerational survival rate of river Wye salmon is around 79%.

$$r = -0.21$$

This is the problem to solve.

- To reverse the decline in population by making r positive requires a circa 20% improvement in intergenerational survival, as a minimum. Less than this, and all we will have achieved is slowing the decline. To get to a healthier, growth regime we would need a 30-40% improvement. If a plan emerges to augment smolt numbers or returning adults, for example, then we need to add 30-40% of what's there naturally. This is the scale of what needs to happen.
- Rather crucially, the negative r problem cannot possibly be addressed by expanding habitat.

K is NOT the problem.

- Also of relevance is the spatial variability of the 1980s fall in parr survival. On a catchment-wide basis, the e-fishing data shows an approximate 2x fall which combined with an approximate 5x fall in marine survival is consistent with the above r value derived from (catchment-wide) rod-catch data. But, at the level of individual e-fishing sites, we found falls of 5x to 12x at some sites, and no fall at others. Thus, remediation measures would need to be appropriately targeted.

References

Green, G. (2026). *An analysis of historical electrofishing surveys from the Wye catchment*.

King, R. A., Toms, S., & Stevens, J. R. (2023). Evaluating the importance of accurate sex ratios on egg deposition targets and conservation limit compliance for Atlantic salmon (*Salmo salar* L.) in the River Tamar, south-west England. *Fisheries Management and Ecology*, 30(2), 161–170.
<https://doi.org/10.1111/fme.12609>